Comparative Analysis Report
Assignment: Hybrid RAG & Fine-Tuning for Customer Support

Stage 4: Fine-Tuning and Evaluation

Task 4.5: Perform Comparative Analysis

4.5.1 Compare All Solution Versions

The three architectures were compared using a common evaluation framework. Generation metrics were calculated on the same reproducible **50-record held-out sample** with SOP-grounded references, while the fine-tuned router was evaluated on the complete **400-record leakage-free test split**. Deterministic inference settings were retained across all versions so that changes could be attributed to retrieval and fine-tuning rather than sampling variation.

The **Baseline** answered directly without retrieval. **Solution V1** used the raw customer query for semantic SOP retrieval, while **Solution V2** used a QLoRA fine-tuned JSON router to predict intent and category before constructing a structured retrieval query.

Comparative Evaluation Results

Metric	Baseline	Solution V1	Solution V2
Execution success	100.00%	100.00%	100.00%
Final-answer format adherence	100.00%	100.00%	100.00%
Functional correctness	54.00%	26.00%	38.00%
Hallucination frequency	46.00%	74.00%	62.00%
Output consistency	100.00%	100.00%	100.00%
ROUGE-1	0.2419	0.3058	0.3285
ROUGE-L	0.1338	0.1708	0.1890
BLEU	0.0123	0.0348	0.0338
Top-1 SOP source agreement	N/A	44.00%	84.00%

N/A: the baseline does not include an SOP retrieval stage.

Comparative Interpretation

Baseline to Solution V1: Adding naive retrieval improved lexical similarity to the SOP-grounded references. ROUGE-1 increased by 26.45%, ROUGE-L by 27.64%, and BLEU by 181.75%. However, top-1 source agreement was only 44.00%, functional correctness fell from 54.00% to 26.00%, and hallucination frequency increased from 46.00% to 74.00%. This shows that retrieving context did not guarantee that the selected policy or the generated answer was correct.

Solution V1 to Solution V2: The fine-tuned router produced the clearest improvement in retrieval. Top-1 SOP source agreement increased from 44.00% to 84.00%, a 40-percentage-point gain and a 90.91% relative increase. The router corrected 20 previously incorrect retrievals and did not degrade any previously correct retrievals in the 50-record sample. ROUGE-1 and ROUGE-L increased by 7.42% and 10.67%, while BLEU decreased slightly by 2.84%.

Overall comparison: Solution V2 achieved the strongest retrieval precision and the highest ROUGE-1 and ROUGE-L scores. It also improved functional correctness by 12 percentage points and reduced hallucination frequency by 12 percentage points relative to Solution V1. Nevertheless, the baseline retained the highest functional-correctness score and the lowest hallucination frequency under the automated audit, so the Hybrid RAG system cannot be considered superior on every metric.

Fine-Tuned Router Evaluation

Router Metric	Records	Solution V2
Strict JSON format adherence	400	100.00%
Exact intent accuracy	400	73.25%
Fuzzy intent accuracy	400	84.75%
Exact category accuracy	400	75.75%
Adversarial exact intent accuracy	68	77.94%
Adversarial fuzzy intent accuracy	68	88.24%

The router produced schema-compliant JSON for every held-out record. Exact intent accuracy remained below the proposed 85% target, but fuzzy accuracy reached 84.75%, indicating that several errors were semantically close label variations. On the 68-record regex-derived adversarial subset, exact and fuzzy accuracies increased to 77.94% and 88.24%, respectively.

Against the proposed success criteria, Solution V2 met the targets for JSON adherence, adversarial intent accuracy, deterministic consistency, and relative ROUGE/BLEU improvement over the baseline. It did not meet the 85% exact-intent target or the hallucination targets.

4.5.2 Document Findings and Recommendations

Key Findings

- Solution V2 was the strongest retrieval architecture: top-1 SOP source agreement reached 84.00%, compared with 44.00% for Solution V1, and it achieved the highest ROUGE-1 and ROUGE-L scores.
- Fine-tuning corrected 20 previously incorrect retrievals and did not degrade any previously correct retrievals in the 50-record sample. Relative to Solution V1, functional correctness improved by 12 percentage points and hallucination frequency fell by 12 percentage points.
- The router produced valid JSON in 100% of cases, but exact intent accuracy of 73.25% and exact category accuracy of 75.75% show that classification errors still affect downstream retrieval.
- All three architectures achieved 100% execution success, final-answer format adherence, and deterministic consistency. However, the baseline remained strongest on the automated functional-correctness and hallucination audit, so improved retrieval did not guarantee complete factual reliability.

Limitations

- Generation metrics were calculated on a reproducible 50-record sample because of Colab inference cost, although router accuracy used the full 400-record held-out split.
- The 68-record adversarial subset was created through regex filtering and is not equivalent to a manually curated adversarial benchmark.
- Functional correctness and hallucination frequency were estimated using rule-based and embedding-similarity thresholds; human reviewers may judge borderline cases differently.
- References were model-generated from oracle SOP context, and top-1 source agreement measured the selected document rather than guaranteeing that every required policy detail was retrieved.

Recommendations

- Improve the router with class-balanced training data, additional examples for confused intent pairs, and contrastive examples that separate semantically similar labels.
- Use constrained JSON decoding and a confidence threshold so uncertain predictions fall back to raw-query retrieval or safe escalation instead of forcing an incorrect route.
- Strengthen retrieval with dense-keyword fusion, metadata filtering, and reranking, then add a post-generation grounding check for timelines, refunds, escalation steps, and contact details.
- Conduct human evaluation of correctness, policy completeness, and unsupported claims, and repeat final synthesis on the complete 400-record test split when compute permits.

Future Work

- Create a manually curated adversarial set containing negation, spelling errors, mixed intents, indirect requests, and emotionally worded complaints.
- Evaluate router-confidence calibration, top-k retrieval, reranking, and source-citation generation, including explicit fallback behaviour for unsupported queries.
- Extend to multi-turn conversations and live order or payment tools only after grounding and refusal behaviour are consistently reliable.

Conclusion

Naive RAG improved lexical similarity to SOP-grounded references, while the fine-tuned router substantially improved SOP selection and further strengthened ROUGE performance. Solution V2 is therefore the most effective retrieval architecture, but its remaining classification errors and elevated hallucination rate show that additional grounding controls and human validation are required before production use.